

Yousef Kart

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Experience

Red Piston

April 2026 – Present

Full Stack Software Developer

Windsor, Ontario

- Built and deployed an HR platform using Next.js, Supabase, and Vercel, implementing authentication, AI-powered employee support, applicant tracking, lead automation, and PTO scheduling
- Rebuilt a legacy Android application using React Native (Expo) and Strapi, expanding support to iOS, and web
- Accelerated the development of an augmented reality platform by converting an experimental feature into a production-ready MVP suitable for customer evaluation

Scelta

February 2025 – September 2025

Software Engineer

Windsor, Ontario

- Led the design and rollout of a shared UI component library used across several production apps, reducing duplicated UI code and accelerating feature delivery
- Built an SSO system enabling seamless authentication across 3 applications, eliminating redundant login flows and improving session security
- Developed a real-time 3D geospatial visualization pipeline for interactive development mapping tools
- Reduced REDMap's initial loading time by 97% via query optimization and endpoint restructuring
- Designed a production-ready Next.js application template that decreased new project build time by more than 50%

Freelance

June 2021 – February 2025

3D Modeler

Remote

- Delivered 3D assets and rendering pipelines for 40+ global clients, optimizing polygon counts and texture workflows to meet real-time performance constraints
- Produced architectural visualizations for \$100M+ developments while reducing asset turnaround times to 3–5 days

Skills

Languages: C++, Python, Java, TypeScript, SQL

Frontend: React, React Native, Next.js, Expo, Tailwind, shadcn/ui

Backend & Database: Node.js, Supabase, Neon, Strapi, PostgreSQL, SQLite, REST APIs

Graphics & Interfaces: OpenGL, ImGui, Qt, WebRTC, Three.js, Mapbox

Tools: CMake, Git, Vercel, OpenCV

Projects

Monte Carlo Simulation Engine | C++20, OpenGL, ImGui

- Built a high-performance Monte Carlo engine to simulate thousands of stochastic asset price paths using GBM
- Developed a real-time OpenGL renderer to visualize path dispersion and terminal payoff distributions
- Implemented interactive ImGui control panel enabling dynamic volatility, drift, timestep, and simulation scaling

Black-Scholes Model Visualizer | C++17, Qt

- Implemented Black-Scholes' pricing engine with real-time Greeks visualization and heat mapping
- Computed option sensitivities (Delta, Gamma, Vega, Theta, Rho) with interactive parameter controls
- Optimized computation pipeline to handle millions of floating-point operations per frame

Eyez and Earz | Python, SQLite, LLMs, Finnhub, Massive, SEC EDGAR, yfinance

- Built an event-driven trading pipeline that ingests real-time financial news and classifies market-moving events
- Engineered a modular data pipeline to process live news feeds and track market snapshots
- Developed a scoring and signal framework combining sentiment analysis, event classification, liquidity metrics, relative volume, and volatility detection to rank high-probability trading setups and reduce false-positives

Education

University of Windsor | 4x Dean's Honour Roll

Bachelor of Computer Science (Honours) & Minor in Mathematics

September 2021 – May 2025

Windsor, Ontario